A synthetic process from identification to validation

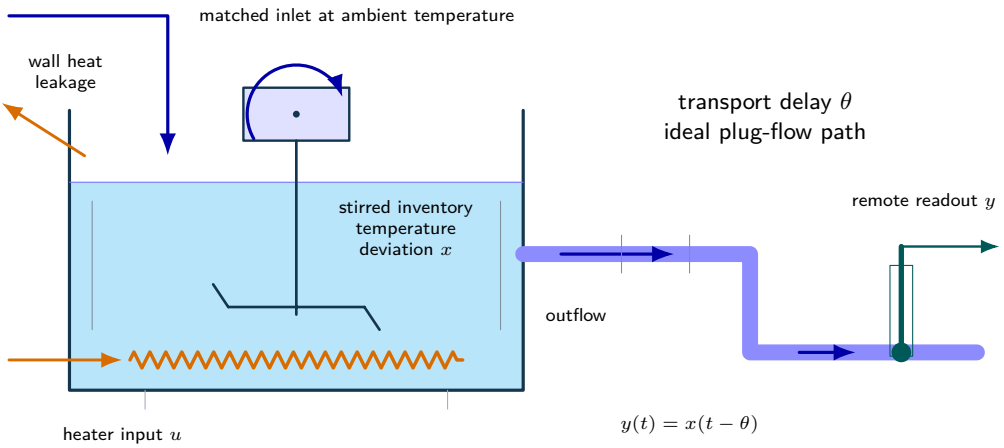
Original learning collection – November 2026

Evidence note. Original teaching scenario. Numerical parameters are deliberately chosen for learning. All generated data and curves are synthetic; no physical measurement is asserted.

1 Design brief

Create a reproducible synthetic identification experiment with gain $K = 1.8$, time constant $\tau = 4\,\mathrm{s}$ and delay $\theta = 0.6\,\mathrm{s}$. Apply a two-unit step at $t = 2\,\mathrm{s}$, sample every $0.05\,\mathrm{s}$ and add independent Gaussian noise with standard deviation 0.015 using seed 2026. Estimate the parameters without initializing at truth, validate on a separate multistep record, and tune PI using $\lambda = 2\,\mathrm{s}$. Assess the delayed closed loop with $u \in [0, 2]$, distinguishing training, validation and simulated controller evaluation.

2 Physical system



One physical interpretation of a first-order-plus-delay model

Synthetic experiment
$G(s) = 1.8e^{-0.6s}/(4s + 1)$. Declared gain, lag and delay generate the records; none is a physical measurement.

Thermal interpretation
One mixed energy store plus ideal transport. Through-flow and wall losses enter one effective conductance.

Identification boundary
Fit on training data, freeze the model, then check a separate multistep record. Geometry does not identify capacity or heater efficiency.

A stirred thermal inventory followed by ideal transport is one possible physical interpretation of the identified transfer function. The effective conductance includes wall leakage and through-flow heat removal. Its fitted gain and time constant do not separately determine capacity, conductance and heater efficiency.

3 A declared synthetic experiment

The teaching plant is $G(s) = Ke^{-\theta s}/(\tau s + 1)$ with $K = 1.8$, $\tau = 4$ s and $\theta = 0.6$ s. These values define a simulation, not measured equipment. For an input step of amplitude A at t_0,

$$y(t) = KA\left(1 - e^{-(t-t_0-\theta)/\tau}\right)\mathbf{1}_{t \geq t_0+\theta}.$$

Independent Gaussian noise is added with a stated seed. The optimizer receives bounds and the starting guess $(1, 2, 0.2)$, not the true parameters.

4 Identification and a separate check

Minimize the sum of squared residuals between the noisy training samples and the delayed-step expression. Use positive bounds on gain and time constant and a nonnegative delay. Then form an independent multistep response by superposing signed step increments. Do not refit on this validation record. Report training and validation RMSE. A good fit on both synthetic records checks the computation under the chosen model class; it does not establish that a real plant follows that class or quantify experimental uncertainty.

5 Controller calculation

With estimated parameters $\hat{K}, \hat{\tau}, \hat{\theta}$ and chosen $\lambda = 2$ s, use the standard IMC-style PI tuning rule

$$K_p = \frac{\hat{\tau}}{\hat{K}(\lambda + \hat{\theta})}, \qquad K_i = K_p/\hat{\tau}.$$

This is the standard APMonitor IMC-style PI tuning rule. Evaluate the controller against the declared true synthetic plant, including its pure input delay and actuator limit.

6 Delay implementation and limitations

Use a 0.01 s numerical control interval. The true delay is exactly 60 such intervals, so a history buffer applies the delayed held input without a Pade approximation. Propagate the first-order plant with its exact held-input update. Add a constant output-equivalent process disturbance after 15 s and use integral back-calculation when the actuator clips. This evaluation has numerical sampling, exact nominal plant parameters and noiseless feedback. It is not a hardware test, robustness proof, statistical confidence interval or validation against independent real measurements.

7 What to learn from the experiment

Training, validation and controller evaluation answer different questions. All three records are synthetic here; experimental claims require independently collected measurements.

8 System sketch

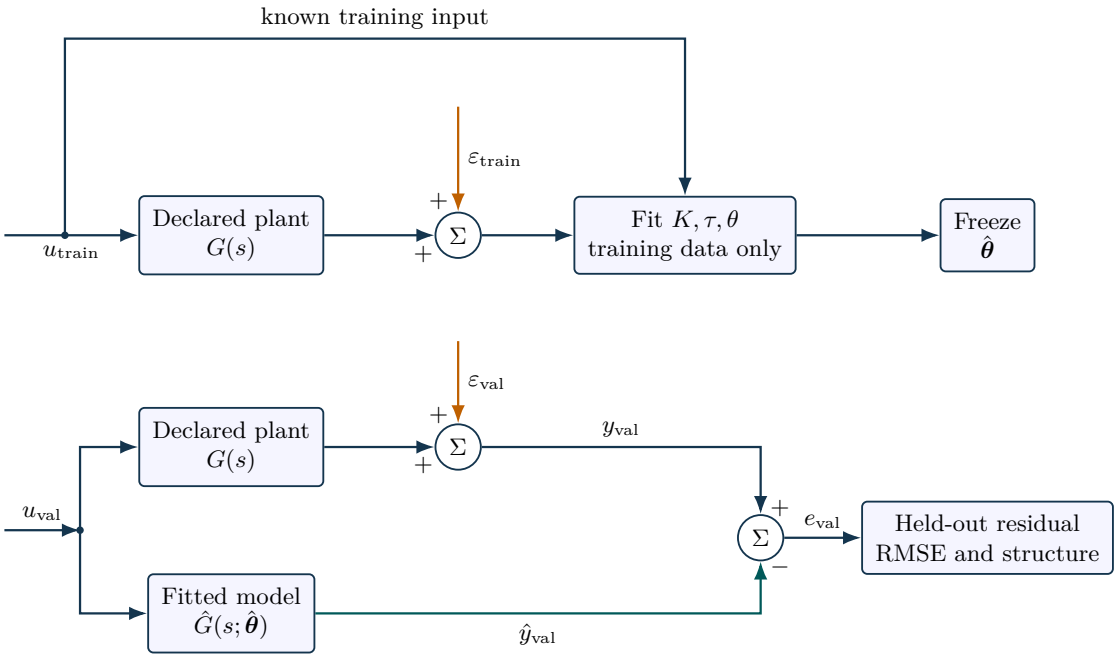
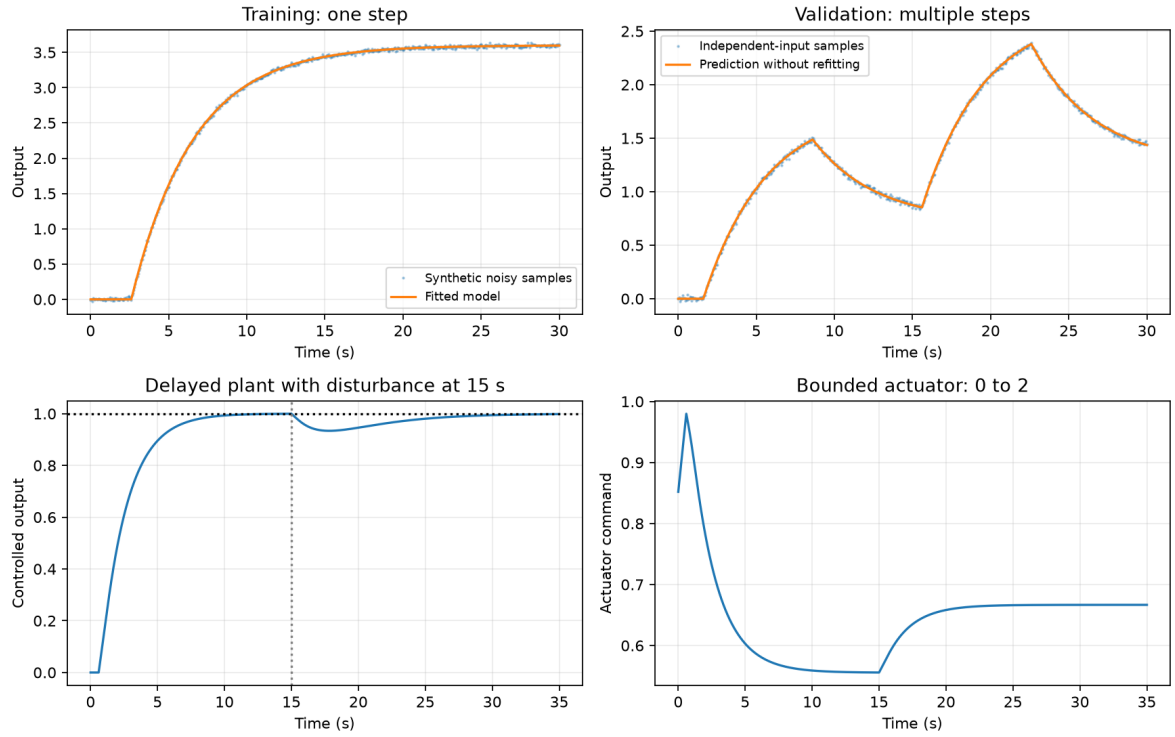


Figure 1. Training selects parameters; held-out validation freezes them. Separate noise records avoid reusing training residuals as validation evidence.

9 Executed computational evidence

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Declared true [gain, tau, delay]: [1.8 4.  0.6]
Estimated [gain, tau, delay]: [1.80002469 3.99349123 0.6042892 ]
Training RMSE=0.015744; independent-input validation RMSE=0.014502
PI: Kp=0.85189303, Ki=0.21332037; final regulated output=0.998318
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Reading the panels. The upper row compares the training fit with prediction on a separate synthetic input record. The lower row shows the simulated closed-loop disturbance response and constrained actuator command. These are numerical checks under the declared model, not measurements of a physical process.